Gabriel Kane

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EDUCATION

Massachusetts Institute of Technology (MIT), Cambridge, MA
Bachelor of Science Mathematics and Music (Double Major)* | *2021

- ***GPA:*** *4.8 out of 5.0*
- ***Relevant Coursework:*** *Real Analysis, Differential Equations, Complex Analysis, Probability and Random Variables, Linear Algebra, Python Programming*

EXPERIENCE

Private Instructor (Virtual & In Person) | 2024 - 2026

- *Delivered personalized instruction in Calculus (AP & college-level), Linear Algebra, Statistics, Algebra I & II, Physics, Python, competition math (AMC), elementary math*
- *Built strong relationships with families through consistent communication, improving student accountability and long-term academic progress*
- *Supported executive functioning skills including organization, study strategies, and test preparation*

Mathematics Faculty, Cambridge School of Weston, Weston, MA | 2021 - 2024

- *Taught and designed high school mathematics courses across multiple levels*
- *Collaborated with faculty on curriculum design, budgeting, and student support decisions*
- *Created problem sets, assessments, and visualizations using LaTeX and Python*

SELECTED PROJECTS

Math Worksheet Generator | 2026 | github.com/gkane1234/elementary_math

- *Built 160,000 line Python question engine and TypeScript app that generate printable Grade 6 through Calculus worksheets across 600+ problem types.*
- *Trained effort models (Pearson $r \approx 0.91$–0.94) so continuous difficulty follows predicted student effort.*

Nonprofit Data Standardization & Dashboarding | 2026

- *Standardized fragmented health datasets using Python and SQL*
- *Built interactive Tableau dashboards to track impact*

Barnes Hut N-Body Simulation | 2025 - 2026 | github.com/gkane1234/gravity

- *Built and optimized ~5000 line simulation using compute shaders*
- *Enabled real-time simulation of 10s of millions of particles through algorithmic optimization*

SKILLS

Teaching: *classroom instruction, one-on-one instruction, curriculum design, assessment development, differentiated learning*
Student Development: *Executive functioning, study skills, academic accountability*
Technical: *Python, Java, SQL, LaTeX, Tableau, Git, Google Data Analytics Certificate | 2026*
Development & Data: *Data analysis, dashboarding, algorithm design*